PRODUCT USER MANUAL

For Multi Observation Sargassum Floating Algae Detection Index

MULTIOBS_GLO_BGC_SURFACE_NRT_015_016
Issue: 1.1

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
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1.1	**27/01/2026**		Declare frozen MODIS based datasets	H.Etienne	H.Etienne

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GLOSSARY AND ABBREVIATIONS

ABI	Advanced Baseline Imager
CAR	Caribbean Sea marine region
CF	Climate Forecast (convention for NetCDF)
CLS	Collecte Localisation Satellites
CMEMS	Copernicus Marine Environment Monitoring Service
DAAC	Distributed Active Archive Center
EROS	Earth Resources Observation and Science center
EU	European Union
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
FAI	Floating Algae Index
FAIA	Floating Algae Index Anomaly
GLO	Global
L1B	Level-1B
L1C	Level-1C
MCI	Maximum Chlorophyll Index
MFC	Monitoring and Forecasting Centre
MOB	Multi Observations
MODIS	Moderate-Resolution Imaging Spectroradiometer
MSI	Multispectral Imager
N/A	Not Applicable

NASA	National Aeronautics and Space Administration
NBS	North Brazil Shelf marine area
NDVI	Normalized Difference Vegetation Index
NetCDF	Network Common Data Form
NFAI	Normalized Floating Algae Index
NOAA	National Oceanic and Atmospheric Administration
NRT	Near Real Time
OLCI	Ocean and Land Color Instrument
OLI	Operational Land Imager
PUM	Product User Manual
QUID	Quality Information Document
SIS	Small Island States
TAC	Thematic Assembly Center
USGS	United States Geological Survey

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download. Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document describes the data product files from the MOB (Multi Observation) **MULTIOBS_GLO_BGC_SURFACE_NRT_015_016** product.
It contains 8 NRT datasets on regular grids from 20 meters to 1.1 kilometer resolution, at a 10-minutes instantaneous, daily and at weekly period, providing instantaneous, aggregated or mean information of the presence of sargassum algae:

- **cmems_obs-mob_car_bgc_nrt_hr_msi_P1D-i**: Near-Real Time daily Normalized Floating Algae Index (NFAI) based on Sentinel-2 MSI observations, Caribbean Sea marine region. Horizontal resolution is 20 meters (0.00018°).
- **cmems_obs-mob_nbs_bgc_nrt_hr_msi_P1D-i:** Near-Real Time daily Normalized Floating Algae Index (NFAI) based on Sentinel-2 MSI observations, North Brazil Shelf marine region. Horizontal resolution is 20 meters (0.00018°).
- **cmems_obs-mob_car_bgc_nrt_hr_oli_P1D-i**: Near-Real Time daily Normalized Floating Algae Index based on Landsat-8 and Landsat-9 OLI observations, Caribbean Sea marine region. Horizontal resolution is 30 meters (0.00025°).
- **cmems_obs-mob_nbs_bgc_nrt_hr_oli_P1D-i**: Near-Real Time daily Normalized Floating Algae Index based on Landsat-8 and Landsat-9 OLI observations, Caribbean Sea marine region. Horizontal resolution is 30 meters (0.00025°).
- **cmems_obs-mob_glo_bgc_nrt_mr_olci_P1D-i:** Near-Real Time daily Normalized Floating Algae Index and Maximum Chlorophyll Index (MCI) based on Sentinel-3 OLCI observations. Horizontal resolution is 275 meters (0.0025°).
- **cmems_obs-mob_glo_bgc_nrt_mr_modis_P1D-i***: Near-Real Time daily Normalized Floating Algae Index based on MODIS Aqua observations. Horizontal resolution is 275 meters (0.0025°).
- **cmems_obs-mob_glo_bgc_nrt_mr_abi_P1D-i:** Near-Real Time 10-minutes instantaneous Floating Algae Index Anomaly (FAIA) based on GOES East ABI observations. Horizontal resolution is 1.1 kilometer (0.01°).
- **cmems_obs-mob_glo_bgc_nrt_mr_multi_P7D-m***: Near-Real Time weekly Normalized Floating Algae Index based on Sentinel-3 OLCI and MODIS Aqua observations, merged. Horizontal resolution is 1.1 kilometer (0.01°).

The NRT 10-minutes instantaneous and daily datasets are updated daily with -1d and -3d information to benefit from the updated NRT input satellite observations.

MULTIOBS_GLO_BGC_SURFACE_NRT_015_016 is only based on satellite observations.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

* Frozen dataset since 23/01/2026

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	**MULTIOBS_GLO_BGC_SURFACE_NRT_015_016**
Geographical coverage	Glo: 100°W – 13°E ; 5°S – 30°N Glo ABI: 100°W – 12°W ; 5°S – 40°N Car: 89°W – 57.5°W; 8°N – 28°N Nbs: 61.5°W – 35°W; 5°S – 10°N
Variables	• Normalized Floating Algae Index • Maximum Chlorophyll Index • Floating Algae Index Anomaly • Normalized Floating Algae Index Weekly Mean • Flag Normalized Floating Algae Index • Flag Maximum Chlorophyll Index • Flag Floating Algae Index Anomaly • Flag Normalized Floating Algae Index Weekly Mean
Product Type	Near Real Time
Available time series	NRT datasets: year-2 to day-1.
Temporal resolution	10-minutes instantaneous- Daily aggregation – Weekly mean
Update Frequency	Daily
Target delivery time	D+1 at 18:30 UTC
Horizontal resolution	20 meters (0.00018°) to 1.1 kilometer (0.01°) on a regular grid
Number of vertical levels	N/A
Format	NetCDF4

NRT runtime schedule:
Every day, daily NRT fields are produced for both the previous day and for three days prior to present.
D+3 is called best analysis and benefits from the last updated satellite observations at level-1, including delayed satellite acquisitions not included in day-1.

b) Production System Description

The methodology used to calculate this product is provided in details in the QUID documentation (CMEMS-MOB-QUID-015-016).

The production system operates 24/7 and is structured as follows:

- Data ingestion: satellite observations and ancillary files are continuously ingested from the different data providers (Copernicus, NASA, NOAA) at level 1 (minimal processing)

- Partial level 2 processing and NFAI/MCI/FAIA calculation: all the satellite images collected are processed to a partial level 2 directly after ingestion. Reflectances are corrected for molecular atmospheric scattering and gaseous absorption only as performing aerosol corrections would filter out the sargassum near infra-red signal. Spectral indices used to detect sargassum are calculated at individual images level, alongside the identification of non-valid pixels (due to lack of satellite coverage, presence of clouds or high sunglint, or land areas).

- Post processing operations are applied to the index values, keeping the index value only for pixels identified as potential sargassum.

- Eventually, the data are aggregated into the final datasets and formatted once a day before delivery.

c) Dataset details

Table 1: list of the datasets and variable names and unit for the MULTIOBS_GLO_BGC_SURFACE_NRT_015_016 product

Product Lines	MULTIOBS_GLO_BGC_SURFACE_NRT_015_016
Geographical coverage	Glo: 100°W – 13°E ; 5°S – 30°N Glo ABI: 100°W – 12°W ; 5°S – 40°N Car: 89°W – 57.5°W; 8°N – 28°N Nbs: 61.5°W – 35°W; 5°S – 10°N
Datasets: cmems_obs-mob_car_bgc_nrt_hr_msi_P1D-i cmems_obs-mob_nbs_bgc_nrt_hr_msi_P1D-i cmems_obs-mob_car_bgc_nrt_hr_oli_P1D-i cmems_obs-mob_nbs_bgc_nrt_hr_oli_P1D-i cmems_obs-mob_glo_bgc_nrt_mr_olci_P1D-i cmems_obs-mob_glo_bgc_nrt_mr_modis_P1D-i**Erreur ! Signet non défini.** cmems_obs-mob_glo_bgc_nrt_mr_abi_P1D-i cmems_obs-mob_glo_bgc_nrt_mr_multi_P7D-m**Erreur ! Signet non défini.**	
Product Type	Near Real Time
Update frequency	Daily
Available time series	-2Y up to D-1
Target delivery time	Daily at 18:30 UTC
Temporal resolution	Daily (aggregated) 10-min instantaneous Weekly mean**Erreur ! Signet non défini.**

Dataset : **cmems_obs-mob_car_bgc_nrt_hr_msi_P1D-i** contains the daily aggregated fields: Normalized Floating Algae Index, Normalized Floating Algae Index Flag. Geographical coverage: 89°W-57.5°W ; 8°N – 28°N
nfai [1] Normalized Floating Algae Index normalized_floating_algae_index
flag_nfai [1] Normalized Floating Algae Index Flag status_flag
Dataset : **cmems_obs-mob_nbs_bgc_nrt_hr_msi_P1D-i** contains the daily aggregated fields: Normalized Floating Algae Index, Normalized Floating Algae Index Flag. Geographical coverage: 61.5°W-35°W ; 5°S – 10.5°N
nfai [1] Normalized Floating Algae Index normalized_floating_algae_index
flag_nfai [1] Normalized Floating Algae Index Flag status_flag
Dataset : **cmems_obs-mob_car_bgc_nrt_hr_oli_P1D-i** contains the daily aggregated fields: Normalized Floating Algae Index, Normalized Floating Algae Index Flag. Geographical coverage: 89°W-57.5°W ; 8°N – 28°N
nfai [1]

Normalized Floating Algae Index normalized_floating_algae_index
flag_nfai [1] Normalized Floating Algae Index Flag status_flag
Dataset : **cmems_obs-mob_nbs_bgc_nrt_hr_oli_P1D-i** contains the daily aggregated fields: Normalized Floating Algae Index, Normalized Floating Algae Index Flag. Geographical coverage: 61.5°W-35°W ; 5°S – 10.5°N
nfai [1] Normalized Floating Algae Index normalized_floating_algae_index
flag_nfai [1] Normalized Floating Algae Index Flag status_flag
Dataset : **cmems_obs-mob_glo_bgc_nrt_mr_olci_P1D-i** contains the daily aggregated fields: Normalized Floating Algae Index, Maximum Chlorophyll Index, Normalized Floating Algae Index Flag, Maximum Chlorophyll Index Flag. Geographical coverage: 100°W-13°E ; 5°S – 30°N
nfai [1] Normalized Floating Algae Index normalized_floating_algae_index
mci [1] Maximum Chlorophyll Index maximum_chlorophyll_index
flag_nfai [1] Normalized Floating Algae Index Flag status_flag
flag_mci [1] Maximum Chlorophyll Index Flag status_flag
Dataset: **cmems_obs-mob_glo_bgc_nrt_mr_modis_P1D-i*** contains the daily aggregated fields: Normalized Floating Algae Index, Normalized Floating Algae Index Flag. Geographical coverage: 100°W-13°E ; 5°S – 30°N
nfai [1] Normalized Floating Algae Index normalized_floating_algae_index
flag_nfai [1] Normalized Floating Algae Index Flag status_flag
Dataset: **cmems_obs-mob_glo_bgc_nrt_mr_abi_P1D-i** contains the 10-minutes instantaneous fields: Floating Algae Index Anomaly, Floating Algae Index Anomaly Flag. Geographical coverage: 100°W-12°W ; 5°S – 40°N
faia [1] Floating Algae Index Anomaly floating_algae_index_anomaly
flag_faia [1] Floating Algae Index Anomaly Flag status_flag
Dataset: **cmems_obs-mob_glo_bgc_nrt_mr_multi_P7D-m***contains the weekly mean fields: Normalized Floating Algae Index Weekly Mean, Normalized Floating Algae Index Weekly Mean Flag.

Geographical coverage: 100°W-13°E ; 5°S – 30°N
nfai [1] Normalized Floating Algae Index normalized_floating_algae_index
flag_nfai [1] Normalized Floating Algae Index Flag status_flag

* Frozen dataset since 23/01/2026

d) Additional information on some parameters

nfai [1]	Normalized Floating Algae Index (NFAI) is calculated from partially corrected reflectance values obtained from satellite observations. The NFAI is a normalized version of the Floating Algae Index (FAI), where the difference between the near-infrared reflectance and a baseline (interpolated from neighboring spectral bands) is divided by the sum of these reflectances. This normalization reduces the variability of the index due to atmospheric conditions and observation geometry, similarly to the approach used for the Normalized Difference Vegetation Index (NDVI) over land surfaces. The NFAI is dimensionless. References Stum et al., 2019; Hu 2009
mci [1]	Maximum Chlorophyll Index (MCI) is calculated from partially corrected reflectance values obtained from satellite observations. The MCI quantifies the height of the radiance peak observed at 709 nm, relative to a baseline between neighboring spectral bands, and is used to indicate areas of high chlorophyll concentration on the water surface. The MCI is dimensionless. Reference Gower et al., 2006
faia [1]	The Floating Algae Index (FAI) anomaly represents an improved version of the FAI as calculated from satellite-derived reflectance data. The FAI anomaly is a dimensionless index. Reference https://hygeos.com/en/project/sargassum_monitoring/

e) Catalogue updating

Every day at D+1, the 10-minutes instantaneous, daily and weekly NRT-analysis are delivered for observations acquired the previous day (D), due to the span of the covered area across the Atlantic from Africa to America.

Every day at D +3, 10-minutes instantaneous, daily and weekly datasets best analyses for D replace the previously delivered NRT-analysis, to mitigate any irregularities in the satellite observations acquisition or delivery process.

The catalogue data from 2-year before present to 3 days before present are not updated unless there are gaps in the data that could not be retrieved and processed in due time.

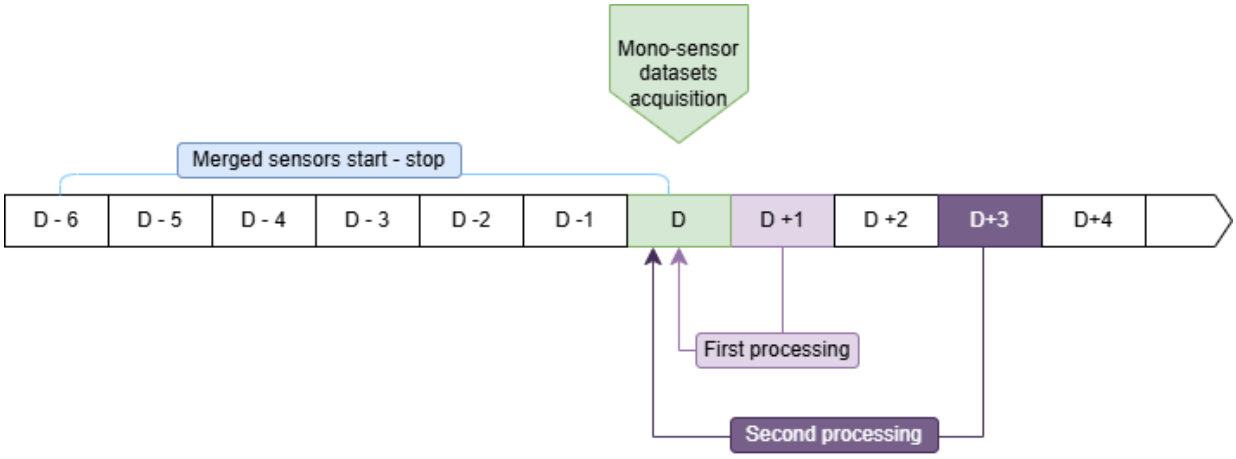

Figure 1 Sargassum NRT production strategy

The Figure 1 shows the timeline for the delivery to the catalogue, with the initial D delivery and update by the D+3 best analysis update.

f) Time averaging

Dataset frequency	Time averaging
weekly	Sliding 7 days (D-1 to D-7) mean value
daily	No averaging: aggregation of all instantaneous satellite observations acquired during a calendar day (local time) over the covered area
10-minutes instantaneous	No averaging: 10-minutes acquisition frame (e.g. : 10:00, 10:10, 10:20, 10:30…)

g) Product use and impact

The MULTIOBS_GLO_BGC_SURFACE_NRT_015_016 product is particularly relevant to the Copernicus Marine Service and the wider oceanographic community as since 2011, an unprecedented volume of sargassum algae (*Sargassum fluitans* and *Sargassum natans*) have been observed in the Gulf of Mexico, Caribbean Sea and West Africa, with massive impact on the lives and livelihoods of local communities when they beach. The region most impacted by these landings is the Caribbean region, home to many Small Island States (SIS) with limited financial and infrastructure resources and in dire need of regular Sargassum observation data in order to set up adequate contingency plans to tackle the algae invasion.

Daily detection product made available to the scientific and nonscientific communities enables institutional stakeholders and private companies to better understand and monitor the movement of the large sargassum raft that affect tourism, commercial shipping, fishing activities, coastal ecosystems and the local population health once they have beached.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

MULTIOBS_GLO_BGC_SURFACE_NRT_015_016 files nomenclature is:
cmems_obs-mob_dom_bgc_nrt_res_captor _ YYYYMMDDThhmmZ.nc

dom: [glo, car, nbs] domain acronym.
res: [hr,mr] resolution acronym.
captor: [msi, oli, olci, modis, abi, multi] captor accronym
YYYYMMDDThhmmZ: date of the product;

Possible file names are listed in Table 2.

5) FILE SIZE

Table 2: MULTIOBS_GLO_BGC_SURFACE_NRT_015_016 file size

DATASET NAME	NAME OF FILE	DIMENSION ESTIMATION [GB]
cmems_obs-mob_car_bgc_nrt_hr_msi_P1D-i	cmems_obs-mob_car_bgc_nrt_hr_msi _YYYYMMDDThhmmZ.nc	13 to 80, to be confirmed depending on compression
cmems_obs-mob_nbs_bgc_nrt_hr_msi_P1D-i	cmems_obs-mob_nbs_bgc_nrt_hr_msi _YYYYMMDDThhmmZ.nc	9 to 51, to be confirmed depending on compression
cmems_obs-mob_car_bgc_nrt_hr_oli_P1D-i	cmems_obs-mob_car_bgc_nrt_hr_oli _YYYYMMDDThhmmZ.nc	6 to 38, to be confirmed

		depending on compression
cmems_obs-mob_nbs_bgc_nrt_hr_oli_P1D-i	cmems_obs-mob_nbs_bgc_nrt_hr_oli_YYYYMMDDThhmmZ.nc	4.5 to 28, to be confirmed depending on compression
cmems_obs-mob_glo_bgc_nrt_mr_olci_P1D-i	cmems_obs-mob_glo_bgc_nrt_mr_olci_YYYYMMDDThhmmZ.nc	0.06
cmems_obs-mob_glo_bgc_nrt_mr_modis_P1D-i*	cmems_obs-mob_glo_bgc_nrt_mr_modis_YYYYMMDDThhmmZ.nc	0.03
cmems_obs-mob_glo_bgc_nrt_mr_abi_P1D-i	cmems_obs-mob_glo_bgc_nrt_mr_abi_YYYYMMDDThhmmZ.nc	0.5
cmems_obs-mob_glo_bgc_nrt_mr_multi_P7D-m*	cmems_obs-mob_glo_bgc_nrt_mr_multi_YYYYMMDDThhmmZ.nc	0.002

* Frozen dataset since 23/01/2026

6) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

7) REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MOB-QUID-015-016.pdf

Gower, J., Hu, C., Borstad, G., King, S., 2006. Ocean Color Satellites Show Extensive Lines of Floating Sargassum in the Gulf of Mexico. IEEE Trans. Geosci. Remote Sens. 44, 3619–3625. https://doi.org/10.1109/TGRS.2006.882258

Hu, C., 2009. A novel ocean color index to detect floating algae in the global oceans. Remote Sens. Environ. 113, 2118–2129. https://doi.org/10.1016/j.rse.2009.05.012

Stum, J., Tebri, H., Sutton, M., Granier, N., 2019. NRT satellite detection and drift forecast of Sargassum algae in the equatorial Atlantic. [Poster Presentation]. 4th International Ocean Color Science meeting, Busan, South Korea, 9-12 April 2019 https://www.aviso.altimetry.fr/fileadmin/documents/data/products/value-added/2019_Poster_SARGASSES_Busan_MSutton_JStum.pdf

8) ANNEX

a) Structure and semantic of NetCDF maps files of dataset cmems_obs-mob_car_bgc_nrt_hr_msi_P1D-i

```
 netcdf cmems_obs-mob_car_bgc_nrt_hr_msi _YYYYMMDDThhmmZ {
dimensions:
        time = 1 ;
        latitude = 111112 ;
        longitude = 175001 ;
variables:
        double time(time) ;
                time:long_name = "Time" ;
                time:standard_name = "time" ;
                time:units = "days since 1950-01-01 00:00:00" ;
                time:calendar = "gregorian" ;
                time:axis = "T" ;
        float latitude(latitude) ;
                latitude:long_name = "Latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
        float longitude(longitude) ;
                longitude:long_name = "Longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
        short nfai(time, latitude, longitude) ;
                nfai:_FillValue = 32767s ;
                nfai:standard_name = "floating_algae_index" ;
                nfai:long_name = "Normalized Floating Algae Index" ;
                nfai:scale_factor = 0.0001 ;
                nfai:units = "1" ;
        short flag_nfai(time, latitude, longitude) ;
                flag_nfai:_FillValue = 32767s ;
                flag_nfai:standard_name = "status_flag” ;
                flag_nfai:long_name = “Normalized Floating Algae Index Flag " ;
                flag_nfai:scale_factor = 1 ;
                flag_nfai:flag_values = 1s, -0.5s, 32767s ;
                flag_nfai:flag_meanings = "no_observations_or_cloud_or_land
algae_free_ocean valid_observation" ;
                flag_nfai:units = "1" ;


// global attributes:
                :title = "Daily sargassum index derived from Sentinel-2 MSI observations for the
Caribbean Sea area" ;
                :source = "Copernicus Sentinel-2 MSI L1C in NRT, provided by Copernicus
        through the Copernicus Dataspace Ecosystem.”
```

```
                :description = " This dataset provides the Normalized Floating Algae Index
(NFAI) derived from satellite observations acquired on a single day (local time) over the
Caribbean Sea region." ;
                :institution = "Collecte Localisation Satellites (CLS)" ;
                :creation_date = "2025-05-21 17:12:29"
                :Conventions = "CF-1.6" ;
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :contact = "sargas-ops@groupcls.com" ;
                :lon_min = -89.0 ;
                :lon_max = -57.5 ;
                :lat_min = 8.0 ;
                :lat_max = 28.0 ;

}
```

b) Structure and semantic of NetCDF maps files of dataset cmems_obs-mob_nbs_bgc_nrt_hr_msi_P1D-i

```
netcdf cmems_obs-mob_nbs_bgc_nrt_hr_msi _YYYYMMDDThhmmZ {
dimensions:
        time = 1 ;
        latitude = 86112 ;
        longitude = 147223 ;
variables:
        double time(time) ;
                time:long_name = "Time" ;
                time:standard_name = "time" ;
                time:units = "days since 1950-01-01 00:00:00" ;
                time:calendar = "gregorian" ;
                time:axis = "T" ;
        float latitude(latitude) ;
                latitude:long_name = "Latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
        float longitude(longitude) ;
                longitude:long_name = "Longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
        short nfai(time, latitude, longitude) ;
                nfai:_FillValue = 32767s ;
                nfai:standard_name = "floating_algae_index" ;
                nfai:long_name = "Normalized Floating Algae Index" ;
                nfai:scale_factor = 0.0001 ;
                nfai:units = "1" ;
        short flag_nfai(time, latitude, longitude) ;
                flag_nfai:_FillValue = 32767s ;
                flag_nfai:standard_name = "status_flag” ;
                flag_nfai:long_name = “Normalized Floating Algae Index Flag" ;
                flag_nfai:scale_factor = 1 ;
                flag_nfai:flag_values = 1s, -0.5s, 32767s ;
                flag_nfai:flag_meanings = "no_observations_or_cloud_or_land
algae_free_ocean valid_observation" ;
                flag_nfai:units = "1" ;


// global attributes:
                :title = "Daily sargassum index derived from Sentinel-2 MSI observations for the
North Brazil Shelf area" ;
                :source = "Copernicus Sentinel-2 MSI L1C in NRT, provided by Copernicus
        through the Copernicus Dataspace Ecosystem.”
                :description = " This dataset provides the Normalized Floating Algae Index
(NFAI) derived from satellite observations acquired on a single day (local time) over the North
Brazil Shelf region." ;
                :institution = "Collecte Localisation Satellites (CLS)" ;
                :creation_date = "2025-05-21 17:12:29"
                :Conventions = "CF-1.6" ;
```

```
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :contact = "sargas-ops@groupcls.com" ;
                :lon_min = -61.5 ;
                :lon_max = -35.0 ;
                :lat_min = -5.0 ;
                :lat_max = 10.5 ;

}
```

c) Structure and semantic of NetCDF maps files of dataset cmems_obs-mob_car_bgc_nrt_hr_oli_P1D-i

```
netcdf cmems_obs-mob_car_bgc_nrt_hr_oli _YYYYMMDDThhmmZ {
dimensions:
        time = 1 ;
        latitude = 80001 ;
        longitude = 126001 ;
variables:
        double time(time) ;
                time:long_name = "Time" ;
                time:standard_name = "time" ;
                time:units = "days since 1950-01-01 00:00:00" ;
                time:calendar = "gregorian" ;
                time:axis = "T" ;
        float latitude(latitude) ;
                latitude:long_name = "Latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
        float longitude(longitude) ;
                longitude:long_name = "Longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
        short nfai(time, latitude, longitude) ;
                nfai:_FillValue = 32767s ;
                nfai:standard_name = "floating_algae_index" ;
                nfai:long_name = "Normalized Floating Algae Index" ;
                nfai:scale_factor = 0.0001 ;
                nfai:units = "1" ;
        short flag_nfai(time, latitude, longitude) ;
                flag_nfai:_FillValue = 32767s ;
                flag_nfai:standard_name = "status_flag” ;
                flag_nfai:long_name = “Normalized Floating Algae Index Flag " ;
                flag_nfai:scale_factor = 1 ;
                flag_nfai:flag_values = 1s, -0.5s, 32767s ;
                flag_nfai:flag_meanings = "no_observations_or_cloud_or_land
algae_free_ocean valid_observation" ;
                flag_nfai:units = "1" ;


// global attributes:
                :title = "Daily sargassum index derived from Landsat-8 and Landsat-9 OLI
observations for the Caribbean Sea area" ;
                :source = "USGS EROS Landsat 8-9 level-1 data in NRT, provided by USGS
        through AWS.”
                :description = " This dataset provides the Normalized Floating Algae Index
(NFAI) derived from satellite observations acquired on a single day (local time) over the
Caribbean Sea region." ;
                :institution = "Collecte Localisation Satellites (CLS)" ;
                :creation_date = "2025-05-21 17:12:29"
                :Conventions = "CF-1.6" ;
```

```
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :contact = "sargas-ops@groupcls.com" ;
                :lon_min = -89.0 ;
                :lon_max = -57.5 ;
                :lat_min = 8.0 ;
                :lat_max = 28.0 ;

}
```

d) Structure and semantic of NetCDF maps files of dataset cmems_obs-mob_nbs_bgc_nrt_hr_oli_P1D-i

```
netcdf cmems_obs-mob_nbs_bgc_nrt_hr_oli _YYYYMMDDThhmmZ {
dimensions:
        time = 1 ;
        latitude = 62001 ;
        longitude = 106001 ;
variables:
        double time(time) ;
                time:long_name = "Time" ;
                time:standard_name = "time" ;
                time:units = "days since 1950-01-01 00:00:00" ;
                time:calendar = "gregorian" ;
                time:axis = "T" ;
        float latitude(latitude) ;
                latitude:long_name = "Latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
        float longitude(longitude) ;
                longitude:long_name = "Longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
        short nfai(time, latitude, longitude) ;
                nfai:_FillValue = 32767s ;
                nfai:standard_name = "floating_algae_index" ;
                nfai:long_name = "Normalized Floating Algae Index" ;
                nfai:scale_factor = 0.0001 ;
                nfai:units = "1" ;
        short flag_nfai(time, latitude, longitude) ;
                flag_nfai:_FillValue = 32767s ;
                flag_nfai:standard_name = "status_flag” ;
                flag_nfai:long_name = “Normalized Floating Algae Index Flag " ;
                flag_nfai:scale_factor = 1 ;
                flag_nfai:flag_values = 1s, -0.5s, 32767s ;
                flag_nfai:flag_meanings = "no_observations_or_cloud_or_land
algae_free_ocean valid_observation" ;
                flag_nfai:units = "1" ;


// global attributes:
                :title = "Daily sargassum index derived from Landsat-8 and Landsat-9 OLI
observations for the North Brazil Shelf area" ;
                :source = " USGS EROS Landsat 8-9 level-1 data in NRT, provided by USGS
        through AWS.”
                :description = " This dataset provides the Normalized Floating Algae Index
(NFAI) derived from satellite observations acquired on a single day (local time) over the North
Brazil Shelf region." ;
                :institution = "Collecte Localisation Satellites (CLS)" ;
```

```
                :creation_date = "2025-05-21 17:12:29"
                :Conventions = "CF-1.6" ;
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :contact = "sargas-ops@groupcls.com";
                :lon_min = -61.5 ;
                :lon_max = -35.0 ;
                :lat_min = -5.0 ;
                :lat_max = 10.5 ;

}
```

e) Structure and semantic of NetCDF maps files of dataset cmems_obs-mob_glo_bgc_nrt_mr_olci_P1D-i

```
netcdf cmems_obs-mob_glo_bgc_nrt_mr_olci _YYYYMMDDThhmmZ {
dimensions:
        time = 1 ;
        latitude = 14001 ;
        longitude = 45201 ;
variables:
        double time(time) ;
                time:long_name = "Time" ;
                time:standard_name = "time" ;
                time:units = "days since 1950-01-01 00:00:00" ;
                time:calendar = "gregorian" ;
                time:axis = "T" ;
        float latitude(latitude) ;
                latitude:long_name = "Latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
        float longitude(longitude) ;
                longitude:long_name = "Longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
        short nfai(time, latitude, longitude) ;
                nfai:_FillValue = 32767s ;
                nfai:standard_name = "floating_algae_index" ;
                nfai:long_name = "Normalized Floating Algae Index" ;
                nfai:scale_factor = 0.0001 ;
                nfai:units = "1" ;
        short mci(time, latitude, longitude) ;
                mci:_FillValue = 32767s ;
                mci:standard_name = "maximum_chlorophyll_index" ;
                mci:long_name = "Maximum Chlorophyll Index" ;
                mci:scale_factor = 0.0001 ;
                mci:units = "1" ;
        short flag_nfai(time, latitude, longitude) ;
                flag_nfai:_FillValue = 32767s ;
                flag_nfai:standard_name = "status_flag” ;
                flag_nfai:long_name = “Normalized Floating Algae Index Flag" ;
                flag_nfai:scale_factor = 1 ;
                flag_nfai:flag_values = 1s, -0.5s, 32767s ;
                flag_nfai:flag_meanings = "no_observations_or_cloud_or_land
algae_free_ocean valid_observation" ;
                flag_nfai:units = "1" ;
        short flag_mci(time, latitude, longitude) ;
                flag_mci:_FillValue = 32767s ;
                flag_mci:standard_name = "status_flag” ;
                flag_mci:long_name = “Maximum Chlorophyll Index Flag" ;
                flag_mci:scale_factor = 1 ;
                flag_mci:flag_values = 1s, -0.5s, 32767s ;
```

```
                flag_mci:flag_meanings = "no_observations_or_cloud_or_land
algae_free_ocean valid_observation" ;
                flag_mci:units = "1" ;


// global attributes:
                :title = "Daily sargassum indices derived from Sentinel-3 OLCI observations" ;
                :source = "Copernicus Sentinel-3 OLCI L1B in NRT, provided by Copernicus
        through the EUMETSAT Data Centre."
                :description = " This dataset provides the Normalized Floating Algae Index
(NFAI) and the Maximum Chlorophyll Index (MCI), both derived from satellite observations
acquired on a single day across the Atlantic Ocean (local time)." ;
                :institution = "Collecte Localisation Satellites (CLS)" ;
                :creation_date = "2025-05-21 17:12:29"
                :Conventions = "CF-1.6" ;
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :contact = "sargas-ops@groupcls.com" ;
                :lon_min = -100 ;
                :lon_max = 13.0 ;
                :lat_min = -5.0 ;
                :lat_max = 30.0 ;


}
```

f) Structure and semantic of NetCDF maps files of dataset cmems_obs-mob_glo_bgc_nrt_mr_modis_P1D-I (Frozen dataset since 23/01/2026)

```
netcdf cmems_obs-mob_glo_bgc_nrt_mr_modis _YYYYMMDDThhmmZ {
dimensions:
        time = 1 ;
        latitude = 14001 ;
        longitude = 45201 ;
variables:
        double time(time) ;
                time:long_name = "Time" ;
                time:standard_name = "time" ;
                time:units = "days since 1950-01-01 00:00:00" ;
                time:calendar = "gregorian" ;
                time:axis = "T" ;
        float latitude(latitude) ;
                latitude:long_name = "Latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
        float longitude(longitude) ;
                longitude:long_name = "Longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
        short nfai(time, latitude, longitude) ;
                nfai:_FillValue = 32767s ;
                nfai:standard_name = "floating_algae_index" ;
                nfai:long_name = "Normalized Floating Algae Index" ;
                nfai:scale_factor = 0.0001 ;
                nfai:units = "1" ;
        short flag_nfai(time, latitude, longitude) ;
                flag_nfai:_FillValue = 32767s ;
                flag_nfai:standard_name = "status_flag” ;
                flag_nfai:long_name = “Normalized Floating Algae Index Flag " ;
                flag_nfai:scale_factor = 1 ;
                flag_nfai:flag_values = 1s, -0.5s, 32767s ;
                flag_nfai:flag_meanings = "no_observations_or_cloud_or_land
algae_free_ocean valid_observation" ;
                flag_nfai:units = "1" ;


// global attributes:
                :title = "Daily sargassum index derived from MODIS Aqua observations" ;
                :source = "Aqua MODIS level-1 data in NRT, provided by NASA, U.S.
        Government, through the Ocean & Biology DAAC.”
                :description = " This dataset provides the Normalized Floating Algae Index
(NFAI) derived from satellite observations acquired on a single day across the Atlantic Ocean
(local time)." ;
                :institution = "Collecte Localisation Satellites (CLS)" ;
                :creation_date = "2025-05-21 17:12:29"
                :Conventions = "CF-1.6" ;
```

```
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :contact = "sargas-ops@groupcls.com" ;
                :lon_min = -100 ;
                :lon_max = 13.0 ;
                :lat_min = -5.0 ;
                :lat_max = 30.0 ;

}
```

g) Structure and semantic of NetCDF maps files of dataset cmems_obs-mob_glo_bgc_nrt_mr_abi_P1D-i

```
netcdf cmems_obs-mob_glo_bgc_nrt_mr_abi _YYYYMMDDThhmmZ {
dimensions:
        time = 60 ;
        latitude = 4501 ;
        longitude = 8801 ;
variables:
        double time(time) ;
                time:long_name = "Time" ;
                time:standard_name = "time" ;
                time:units = "days since 1950-01-01 00:00:00" ;
                time:calendar = "gregorian" ;
                time:axis = "T" ;
        float latitude(latitude) ;
                latitude:long_name = "Latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
        float longitude(longitude) ;
                longitude:long_name = "Longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
        short faia(time, latitude, longitude) ;
                faia:_FillValue = 32767s ;
                faia:standard_name = "floating_algae_index" ;
                faia:long_name = "Floating Algae Index Anomaly" ;
                faia:scale_factor = 0.0001 ;
                faia:units = "1" ;
        short flag_faia(time, latitude, longitude) ;
                flag_faia:_FillValue = 32767s ;
                flag_faia:standard_name = "status_flag” ;
                flag_faia:long_name = “Floating Algae Index Anomaly Flag" ;
                flag_faia:scale_factor = 1 ;
                flag_faia:flag_values = 1s, -0.5s, 32767s ;
                flag_faia:flag_meanings = "no_observations_or_cloud_or_land
algae_free_ocean valid_observation" ;
                flag_faia:units = "1" ;


// global attributes:
                :title = "10-minutes instantaneous sargassum index derived from GOES-East
ABI observations" ;
                :source = "GOES-East ABI level 1 data in NRT, provided by NOAA and
        distributed by SATMOS (Meteo-France / Centre de Meteorologie Spatiale) -
        AERIS/ICARE.”
                :description = " This dataset provides the instantaneous Floating Algae Index
Anomaly derived from satellite observations acquired every 10 minutes between 10:00 UTC
and 19:50 UTC across the Atlantic Ocean.” ;
                :institution = "Collecte Localisation Satellites (CLS)" ;
                :creation_date = "2025-05-21 17:12:29"
```

```
                :Conventions = "CF-1.6" ;
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :contact = "sargas-ops@groupcls.com" ;
                :lon_min = -100 ;
                :lon_max = -12.0 ;
                :lat_min = -5.0 ;
                :lat_max = 40.0 ;

}
```

h) Structure and semantic of NetCDF maps files of dataset of cmems_obs-mob_glo_bgc_nrt_mr_multi_P7D-m (Frozen dataset since 23/01/2026)

```
netcdf cmems_obs-mob_glo_bgc_nrt_mr_multi _YYYYMMDDThhmmZ {
dimensions:
        time = 1 ;
        latitude = 3500 ;
        longitude = 11300 ;
variables:
        double time(time) ;
                time:long_name = "Time" ;
                time:standard_name = "time" ;
                time:units = "days since 1950-01-01 00:00:00" ;
                time:calendar = "gregorian" ;
                time:axis = "T" ;
        float latitude(latitude) ;
                latitude:long_name = "Latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:axis = "Y" ;
        float longitude(longitude) ;
                longitude:long_name = "Longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:axis = "X" ;
        short nfai(time, latitude, longitude) ;
                nfai:_FillValue = 32767s ;
                nfai:standard_name = "floating_algae_index" ;
                nfai:long_name = " Normalized Floating Algae Index Weekly Mean" ;
                nfai:scale_factor = 0.0001 ;
                nfai:units = "1" ;
        short flag_nfai(time, latitude, longitude) ;
                flag_nfai:_FillValue = 32767s ;
                flag_nfai:standard_name = "status_flag” ;
                flag_nfai:long_name = “Normalized Floating Algae Index Flag" ;
                flag_nfai:scale_factor = 1 ;
                flag_nfai:flag_values = 1s, -0.5s, 32767s ;
                flag_nfai:flag_meanings = "no_observations_or_cloud_or_land
algae_free_ocean valid_observation" ;
                flag_nfai:units = "1" ;


// global attributes:
                :title = "Weekly mean sargassum index derived from Sentinel-3 OLCI and
MODIS Aqua observations" ;
                :source = " Copernicus Sentinel-3 OLCI L1B in NRT, provided by Copernicus
        through the EUMETSAT Data Centre. Aqua MODIS level-1 data in NRT, provided by
        NASA, U.S. Government, through the Ocean & Biology DAAC.”
                :description = " This dataset provides weekly mean statistics of the Normalized
Floating Algae Index (NFAI) derived from satellite observations acquired on the last 7 days
across the Atlantic Ocean.”;
                :institution = "Collecte Localisation Satellites (CLS)" ;
                :creation_date = "2025-05-21 17:12:29"
```

```
                :Conventions = "CF-1.6" ;
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :contact = "sargas-ops@groupcls.com" ;
                :lon_min = -100 ;
                :lon_max = 13.0 ;
                :lat_min = -5.0 ;
                :lat_max = 30.0 ;

}
```